

Title of Your Paper

Project Report for EN.553.472/672.SP24 Graph Theory

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Abstract

This is the abstract of your paper, which should briefly summarize the problem you are addressing, your approach, and the main results. It should be self-contained and readable for a wide audience.

You can find some useful latex tutorials here:

<https://www.overleaf.com/learn/latex/Tutorials>

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1 Introduction

The introduction should briefly place the work in a broad context and highlight why it is important. It should define the problem under consideration and the approach or solution proposed. Cite relevant references [1], [4], [2], [3] to support your statements.

2 Main Results

Theorem 2.1 (Ramsey Theorem). *For all positive integers r, s , there exists a positive integer $R(r, s)$ such that for any $N \geq R(r, s)$ every graph G with N vertices satisfies $\alpha(G) \geq r$ or $\alpha(G) \geq s$.*

Proof. Provide the proof of your theorem here. □

Lemma 2.2.

$$R(r, s) \leq R(r - 1, s) + R(r, s - 1). \quad (1)$$

3 Aligned Equations

Example of a single equation number:

$$\begin{aligned} \vec{x}^T(I_n + \mathcal{A})\vec{x} &= \vec{x}^T\vec{x} + \vec{x}^T\mathcal{A}\vec{x} \\ &= \sum_{i=1}^n x_i^2 + \sum_{v_i v_j \in E(G)} \frac{2x_i x_j}{\sqrt{d(v_i)}\sqrt{d(v_j)}} \\ &= \sum_{v_i v_j \in E(G)} \frac{x_i^2}{d(v_i)} + \frac{x_j^2}{d(v_j)} + \sum_{v_i v_j \in E(G)} \frac{2x_i x_j}{\sqrt{d(v_i)}\sqrt{d(v_j)}} \quad (2) \\ &= \sum_{v_i v_j \in E(G)} \left(\frac{x_i}{\sqrt{d(v_i)}} + \frac{x_j}{\sqrt{d(v_j)}} \right)^2 \\ &\geq 0. \end{aligned}$$

This (2) is how you reference an equation.

Example where every line gets a separate equation number:

$$\vec{x}^T(I_n + \mathcal{A})\vec{x} = \vec{x}^T\vec{x} + \vec{x}^T\mathcal{A}\vec{x} \quad (3)$$

$$= \sum_{i=1}^n x_i^2 + \sum_{v_i v_j \in E(G)} \frac{2x_i x_j}{\sqrt{d(v_i)}\sqrt{d(v_j)}} \quad (4)$$

$$= \sum_{v_i v_j \in E(G)} \frac{x_i^2}{d(v_i)} + \frac{x_j^2}{d(v_j)} + \sum_{v_i v_j \in E(G)} \frac{2x_i x_j}{\sqrt{d(v_i)}\sqrt{d(v_j)}} \quad (5)$$

$$= \sum_{v_i v_j \in E(G)} \left(\frac{x_i}{\sqrt{d(v_i)}} + \frac{x_j}{\sqrt{d(v_j)}} \right)^2 \quad (6)$$

$$\geq 0. \quad (7)$$

This (5) is how you reference a particular line in your equation.

Example where equation numbers are skipped all together:

$$\vec{x}^T(I_n + \mathcal{A})\vec{x} = \vec{x}^T\vec{x} + \vec{x}^T\mathcal{A}\vec{x}$$

$$= \sum_{i=1}^n x_i^2 + \sum_{v_i v_j \in E(G)} \frac{2x_i x_j}{\sqrt{d(v_i)}\sqrt{d(v_j)}}$$

$$= \sum_{v_i v_j \in E(G)} \frac{x_i^2}{d(v_i)} + \frac{x_j^2}{d(v_j)} + \sum_{v_i v_j \in E(G)} \frac{2x_i x_j}{\sqrt{d(v_i)}\sqrt{d(v_j)}}$$

$$= \sum_{v_i v_j \in E(G)} \left(\frac{x_i}{\sqrt{d(v_i)}} + \frac{x_j}{\sqrt{d(v_j)}} \right)^2$$

$$\geq 0.$$

4 Images

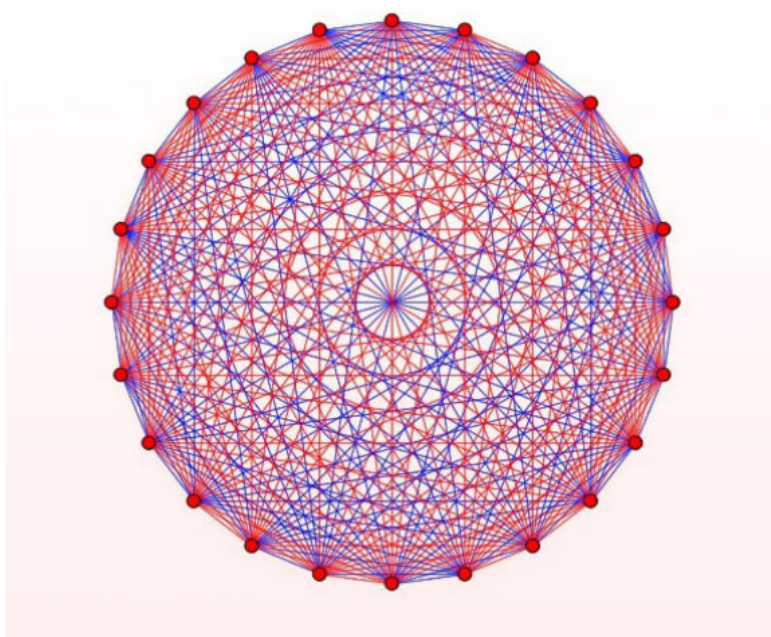


Figure 1: Example of a graph

You can reference the figure as Figure [1](#).

5 Conclusion

Summarize the main findings and their importance, and state the future work directions.

References

- [1] A. Author and B. Another. Title of the article. *Journal Name*, 42(3):123–456, 2024.
- [2] D. Contributor. Title of the conference paper. In *Proceedings of the Conference*, pages 789–1011, Conference Location, 2024. Organization Name.

- [3] E. Facilitator. Title of the misc reference. How it was published, 2024.
Additional Information.
- [4] C. Writer. *Title of the Book*. Publisher Name, City, Country, 2023.